Shivam Juyal

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Summary

Software engineer building **LLM systems** and backend services, with a focus on evaluating their **reliability**. Skilled in **RAG**, **embeddings-based** retrieval, APIs, and databases using **Python**, **TypeScript**, **Node.js** and **PostgreSQL**.

Experience

Webrex Studio Jan 2026 – Mar 2026
MERN Developer Intern *Ahmedabad, Remote*

- Built the **graph-generation** step of a **Claude Code slash command** that models a codebase as a **graph of entities and relationships** (files as nodes with short summaries, linked by edges), and **integrated** the team's existing **hybrid semantic + keyword RAG pipeline** to retrieve relevant nodes so Claude reads only those files and their neighbors, not the whole repo.
- Enriched user prompts with **graph-retrieved** context in a single pass, cutting back-and-forth with Claude; enrichment completes in **under 15 seconds** on a **500-file** repo.
- Built an **incremental, event-driven daemon** replacing full graph rebuilds on every significant change; it re-processes only changed files, cutting rebuild compute and LLM token usage an **estimated 75–80%** (typical changes touched 20–25% of files).
- Resolved production issues across **REST APIs**, **Node.js** services, and **MongoDB** by tracing logs to root cause.

Projects

Quill | *Python, Next.js, Node.js, PostgreSQL, Redis, BullMQ, Groq, Gemini, Cohere Reranker* GitHub | Live Demo

- Built a RAG PDF Q&A app with **embeddings-based pgvector** retrieval, **async BullMQ/Redis ingestion**, and **Groq-to-Gemini failover** for rate limits and outages; added **GitHub Actions CI** running server tests on every push and pull request.
- Evaluated retrieval on a 50-question golden set: hybrid vector + keyword **RRF** with **Cohere re-ranking** raised **hit@1 from 54% (vector-only) to 80%** and **MRR@8 from 0.687 to 0.857**.
- Scored 50 answers with LLM-as-Judge (**86.7% correctness, 92.7% faithfulness, 87.7% relevancy**).

Inflow | *TypeScript, Node.js, React, LangGraph, Groq AI, Gmail/Calendar API, Supabase, Zod, Vitest* GitHub

- Built an AI **email-automation** agent with **LangGraph** that classifies Gmail into 6 categories, evaluated at **88%** accuracy on **200** emails, using Zod-validated **structured** LLM output and OAuth 2.0.
- Designed a **human-in-the-loop** flow (drafts and meetings only after approval), with **idempotent processing** to prevent duplicate replies; wrote **175 automated Vitest tests** to verify it.

PushCloud | *React, Node.js, Docker, AWS (ECS Fargate, S3), Redis, Socket.io, MinIO, Supabase, Vitest* GitHub | Demo Video

- Built a multi-tenant platform serving React/Vite apps via **reverse-proxy** routing; ran on **AWS (ECS Fargate, S3)**, then migrated to **Docker** for lower cost and control; load-tested **10 concurrent builds**, passing **18/18 checks**.
- Added tenant-isolated **Socket.io** build-log streaming; wrote **Vitest** security tests that blocked **39/39 unauthorized attempts and 44/44 injection payloads**, with no log loss or reordering across **700 logs**.

Technical Skills

Languages: Python, TypeScript, SQL, C++
AI & LLM: RAG, Embeddings, Vector DBs (pgvector), Structured LLM Output, LangGraph, LLM APIs (Groq, Gemini), Graph-based Retrieval
Backend & Data: Node.js, REST APIs, PostgreSQL, MongoDB, Redis
Testing & DevOps: Vitest, GitHub Actions (CI/CD), Docker, AWS

Education

Graphic Era Hill University Dehradun, Uttarakhand
Bachelor of Technology in Computer Science and Engineering *June 2021 – July 2025*

Achievements

Built and published **@the-white-wolf/ai-commit** to npm, a CLI that generates commit messages from diffs, reaching **1800+** downloads.
Solved **1,000+** Data Structures and Algorithms problems across competitive programming platforms; reached a peak **LeetCode** rating of **1,700+**.